

Total No. of Questions: 8]

SEAT No. :

PA-991

[5927]-460

[Total No. of Pages : 2

B.E. (Honors in Robotics)
INDUSTRIAL ROBOTICS AND AUTOMATION
(2019 Pattern) (Semester-VII) (404181)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2 and Q3 or Q4 and Q5 or Q6 and Q7 or Q8.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) What is pneumatic system? List the industrial applications of it and elaborate any one. [6]
- b) List different methods to control pneumatic control system. What is Karnaugh - Veitch mapping? [6]
- c) Enlist the basic components in pneumatic system and explain them in detail. [6]

OR

- Q2)** a) Draw different types of valve symbols and explain any one with the example. [6]
- b) List the advantages and disadvantages of pneumatic system. [6]
- c) Draw and explain single acting and double acting cylindrical actuators with proper operation. [6]

- Q3)** a) Write the steps to design of parts for high speed feeding and orienting. State different feeding difficulties. [9]
- b) How to perform the analysis of an assembly. Explain with one example. [8]

OR

- Q4)** a) Explain in detail how high speed automatic insertion is possible in robotic assembly. State general rules for product design and automation. [9]
- b) State design features of CNC systems. Explain drive system for CNC machine tools. [6]

P.T.O.

- Q5)** a) Define Mechatronics. Explain different key elements of the Mechatronics system. [6]
b) State and explain different stages in designing a Mechatronics system. [6]
c) Write a case study on Pick and Place robot. [6]

OR

- Q6)** a) Give the classification of a Mechatronics System. [6]
b) How mechatronics system design is different than traditional system? [6]
c) Write a case study on engine management system. [6]
- Q7)** a) Classify the pumps. Give characteristics and selection criteria for Pumps. [9]
b) With an example explain the calculations for power and velocity during extension and retraction. [8]

OR

- Q8)** a) With a suitable diagram and PLC ladder diagram explain how to control two cylinders? [9]
b) Classify the control valves used in hydraulic system. Explain pressure control valves in detail with its functions. [8]

